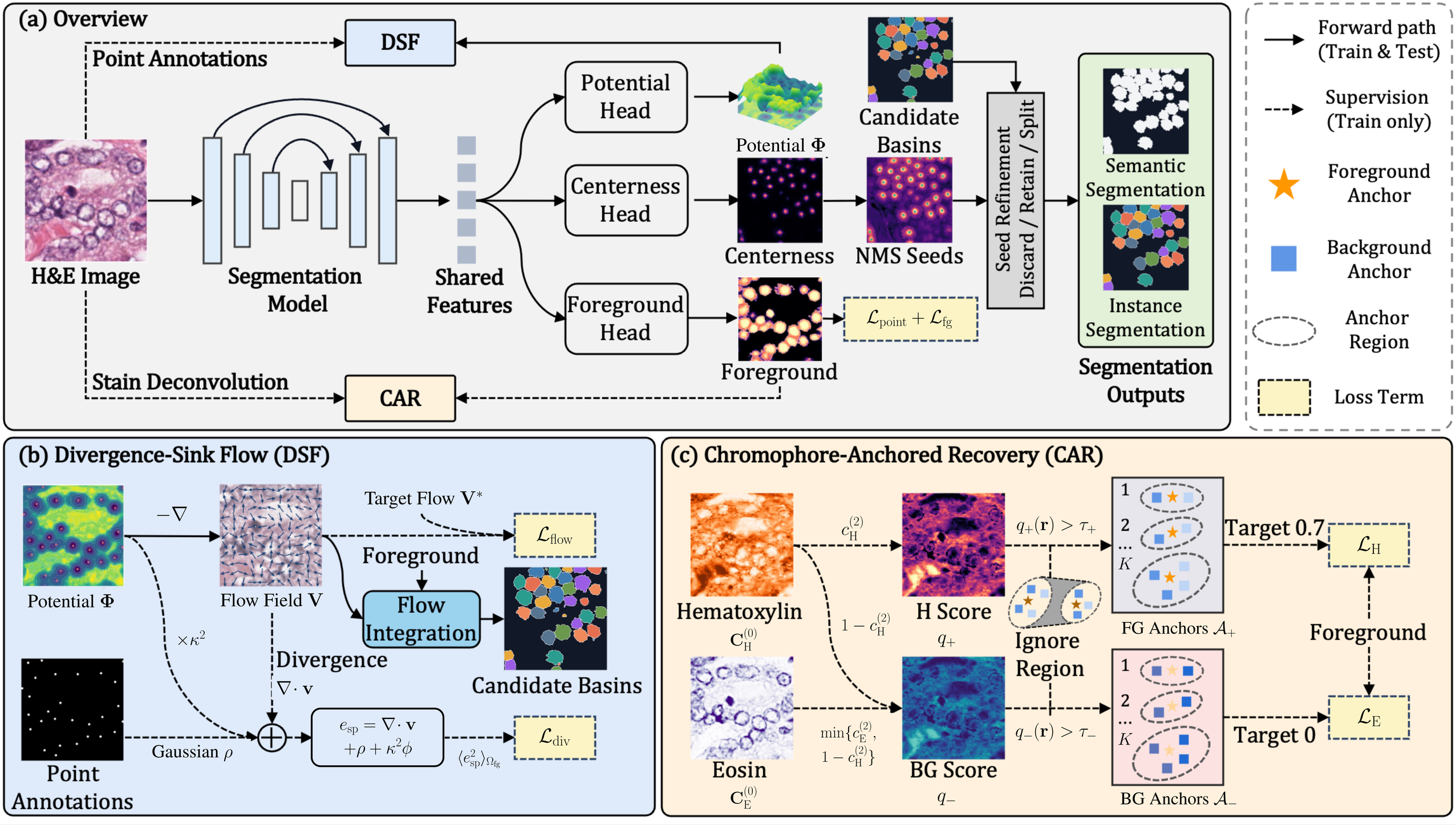

(a) Overview
Point Annotations
DSF
H&E Image
Segmentation Model
Shared Features
Potential Head
Centerness Head
Foreground Head
Potential Φ
Centerness
Foreground
Candidate Basins
NMS Seeds
Seed Refinement
Discard / Retain / Split
Semantic Segmentation
Instance Segmentation
Segmentation Outputs
$\mathcal{L}_{\text{point}} + \mathcal{L}_{\text{fg}}$
Stain Deconvolution
CAR
Forward path (Train & Test)
Supervision (Train only)
Foreground Anchor
Background Anchor
Anchor Region
Loss Term
(b) Divergence-Sink Flow (DSF)
Potential Φ
$-\nabla$
Flow Field V
Target Flow $\mathbf{V}^*$
$\mathcal{L}_{\text{flow}}$
Foreground
Flow Integration
Candidate Basins
$\times\kappa^2$
Divergence
$\nabla \cdot \mathbf{v}$
Point Annotations
Gaussian $\rho$
$e_{\text{sp}} = \nabla \cdot \mathbf{v} + \rho + \kappa^2 \phi$
$\langle e_{\text{sp}}^2 \rangle_{\Omega_{\text{fg}}}$
$\mathcal{L}_{\text{div}}$
(c) Chromophore-Anchored Recovery (CAR)
Hematoxylin
$\mathbf{C}_{\text{H}}^{(0)}$
Eosin
$\mathbf{C}_{\text{E}}^{(0)}$
$c_{\text{H}}^{(2)}$
$1 - c_{\text{H}}^{(2)}$
$\min\{c_{\text{E}}^{(2)}, 1 - c_{\text{H}}^{(2)}\}$
H Score
$q_+$
BG Score
$q_-$
$q_+(\mathbf{r}) > \tau_+$
$q_-(\mathbf{r}) > \tau_-$
Ignore Region
FG Anchors $\mathcal{A}_+$
BG Anchors $\mathcal{A}_-$
Target 0.7
Target 0
$\mathcal{L}_{\text{H}}$
$\mathcal{L}_{\text{E}}$
Foreground